
Physics-Informed Gaussian Processes for Spatiotemporal Wind Field Estimation

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Abstract

Wind is a continuous vector field critical for applications such as weather forecasting, wind energy assessment, and UAV-based cloud seeding operations. Reconstructing the full wind field from sparse station observations remains a challenging problem, as classical numerical weather prediction is computationally expensive while data-driven methods lack spatial uncertainty quantification. This project proposes a two-stage pipeline combining Gradient Boosted Trees (GBT) with a physics-informed Gaussian Process (GP) to reconstruct spatiotemporal wind fields at 850hPa from sparse observations. The GBT captures nonlinear relationships between meteorological features and wind at station locations, while the GP interpolates spatially using a spatiotemporal kernel encoding physical priors. Evaluated on ERA5 reanalysis data over southern California, the proposed method achieves a 33% RMSE reduction over GBT alone and 91% uncertainty coverage, significantly outperforming spatial average and kriging baselines. Results demonstrate that spatiotemporal GP regression with learned hyperparameters provides accurate, well-calibrated wind field estimates suitable for cloud seeding mission planning. Track: Empirical.

1 Introduction

Accurate wind field estimation is a fundamental problem in atmospheric science with direct relevance to weather forecasting, wind energy siting, and the emerging application of UAV-based cloud seeding. Cloud seeding operations, in which drones disperse ice-nucleating agents into cloud layers at altitudes of 1500-3000m, require precise knowledge of the wind field at those altitudes to plan safe and effective flight paths. In practice, however, wind measurements are sometimes only available at a sparse, irregular set of observation stations or radiosonde launch sites. Reconstructing the full continuous wind field from these sparse observations, as well as quantifying uncertainty in that reconstruction, is therefore a problem of significant practical importance.

Classical approaches to this problem fall into two categories. Numerical weather prediction (NWP) methods such as ECMWF's IFS solve the governing fluid dynamics equations forward in time on a dense grid. While physically rigorous, these methods require substantial computational resources and involve many hand-tuned parameterizations, making them impractical for real-time UAV mission planning. Data-driven methods, including gradient boosted trees and deep learning, have shown strong performance at predicting wind at fixed station locations from meteorological features, but they produce point estimates with no sense of spatial uncertainty, which can be a critical limitation for safety-sensitive applications.

Gaussian process (GP) regression offers a principled middle ground. GPs produce probabilistic predictions with calibrated uncertainty estimates, and their covariance structure can encode physical priors about wind behavior through kernel design. However, standard GP approaches do not leverage the rich feature information available at station locations, limiting their predictive accuracy.

This project proposes a two-stage pipeline that combines the complementary strengths of gradient boosted trees and physics-informed GP regression for spatiotemporal wind field estimation at 850hPa. The pipeline is evaluated on ERA5 reanalysis data over southern California (a globally consistent gridded dataset produced by ECMWF that assimilates historical observations into a physics model to provide complete atmospheric fields at 0.25° resolution), making it a natural ground truth for sparse interpolation experiments. The GBT stage learns nonlinear mappings from local meteorological features to wind speed, while the GP stage interpolates the residuals spatially using a spatiotemporal kernel. The pipeline is evaluated on ERA5 reanalysis data over southern California, a region with complex terrain and strong mesoscale wind variability relevant to cloud seeding operations. The main contributions of this project are: (1) a two-stage GBT + GP pipeline for spatiotemporal wind field reconstruction from sparse observations; (2) a comparison of three GP kernel designs including a physics-informed divergence-free kernel; and (3) empirical evaluation demonstrating 33% RMSE reduction over GBT alone and 91% uncertainty coverage on held-out ERA5 grid points.

2 Related Work

3 Related Work

Geostatistical interpolation. Kriging, the classical geostatistical method for spatial interpolation, has long been applied to atmospheric field estimation Cressie [1993]. Ordinary kriging fits a variogram to observed data and produces weighted interpolations at unobserved locations. While effective for purely spatial problems, kriging is a linear method that does not incorporate auxiliary meteorological features and provides only simple error estimates rather than full posterior distributions. The proposed pipeline uses kriging as a baseline to contextualize the improvement offered by the GP formulation.

Physics-informed machine learning. Physics-informed neural networks (PINNs) introduced a framework for embedding physical constraints described by partial differential equations directly into the training of neural networks Raissi et al. [2019]. This work draws on the same philosophy of encoding physical priors into a data-driven model, but does so through kernel design in a Gaussian process rather than through PDE residuals in a neural network. The divergence-free kernel used in Stage 2 enforces the incompressibility constraint $\nabla \cdot \mathbf{u} = 0$ directly in the covariance structure, without requiring additional loss terms or post-hoc enforcement Macêdo and Castro [2010].

Gaussian process regression. Gaussian processes provide a principled probabilistic framework for spatial interpolation with calibrated uncertainty Rasmussen and Williams [2006]. Standard scalar GP regression has been widely applied to meteorological field estimation, but treating wind components independently ignores the physical coupling between u and v . Vector-valued GP kernels, including divergence-free and curl-free constructions, address this by modeling both components jointly Cressie [1993]. This work applies a spatiotemporal extension of the divergence-free kernel and compares it against standard scalar kernels on a real atmospheric dataset.

Gradient boosted trees for meteorological prediction. Gradient boosted trees have been widely applied to wind speed forecasting from station data, demonstrating strong performance on tabular meteorological features with nonlinear interactions Friedman [2001]. However, existing applications are generally aimed at point forecasts at fixed stations and do not address spatial interpolation or uncertainty quantification. The proposed pipeline incorporates GBT as the first stage to leverage its strengths on tabular data, while delegating spatial reasoning and uncertainty quantification to the GP stage.

4 Method

4.1 Problem Formulation

Let $\mathcal{S} = \{(\mathbf{x}_i, t_i)\}_{i=1}^N$ denote a sparse set of observation locations, where $\mathbf{x}_i = (\text{lat}_i, \text{lon}_i)$ is a spatial coordinate and t_i is a timestamp. At each observation, the wind vector (u_i, v_i) is measured, where u and v denote the eastward and northward wind components respectively. The goal is to reconstruct the full wind field $\mathbf{f}(\mathbf{x}, t) = (u(\mathbf{x}, t), v(\mathbf{x}, t))$ at all unobserved grid points, along with calibrated uncertainty estimates.

4.2 Stage 1: Gradient Boosted Trees

The first stage learns a mapping from local meteorological features to wind speed at each station location. For each observation i , a feature vector is constructed as:

$$\phi_i = \left[\text{lat}_i, \text{lon}_i, T_i, z_i, \sin\left(\frac{2\pi h_i}{24}\right), \cos\left(\frac{2\pi h_i}{24}\right) \right] \quad (1)$$

where T_i is temperature, z_i is geopotential, and h_i is the hour of day encoded cyclically to preserve periodicity. Two separate GBT models are trained, one for u and one for v :

$$\hat{u}_i = f_u(\phi_i), \quad \hat{v}_i = f_v(\phi_i) \quad (2)$$

GBT is chosen over linear regression and neural networks for its strong performance on small tabular meteorological datasets with nonlinear feature interactions. Ma et al. [2018] With only 100 station locations, neural networks would be prone to overfitting, while linear models cannot capture the nonlinear dependence of wind on elevation and temperature. The residuals from Stage 1 are computed as:

$$r_i^u = u_i - \hat{u}_i, \quad r_i^v = v_i - \hat{v}_i \quad (3)$$

These residuals represent the spatially structured component of wind variability that local features alone cannot explain, and are passed as targets to Stage 2.

4.3 Stage 2: Physics-Informed Gaussian Process

The second stage fits a spatiotemporal Gaussian process to the residuals $\{r_i^u, r_i^v\}$ at the station locations and interpolates across the full grid. A Gaussian process is a distribution over functions fully specified by a mean function and a covariance kernel. The mean function is set to zero since the residuals are approximately mean-centered, and the kernel encodes both spatial and temporal correlation structure.

4.3.1 Spatiotemporal Kernel

A separable spatiotemporal kernel is used:

$$k((\mathbf{x}, t), (\mathbf{x}', t')) = k_{\text{spatial}}(\mathbf{x}, \mathbf{x}') \cdot k_{\text{temporal}}(t, t') \quad (4)$$

where k_{temporal} is a squared exponential kernel with temporal length scale ℓ_t :

$$k_{\text{temporal}}(t, t') = \exp\left(-\frac{(t - t')^2}{2\ell_t^2}\right) \quad (5)$$

and k_{spatial} is one of three candidate kernels compared in this work.

4.3.2 Kernel Designs

RBF kernel. The squared exponential kernel assumes infinite smoothness and models u and v independently:

$$k_{\text{RBF}}(\mathbf{x}, \mathbf{x}') = \sigma^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\ell_s^2}\right) \quad (6)$$

Matérn 5/2 kernel. Relaxes the smoothness assumption to allow the field to be twice differentiable, better matching the physical roughness of atmospheric wind:

$$k_{\text{Mat}}(\mathbf{x}, \mathbf{x}') = \sigma^2 \left(1 + \frac{\sqrt{5}r}{\ell_s} + \frac{5r^2}{3\ell_s^2}\right) \exp\left(-\frac{\sqrt{5}r}{\ell_s}\right) \quad (7)$$

where $r = \|\mathbf{x} - \mathbf{x}'\|$.

Divergence-free kernel. A matrix-valued kernel that models u and v jointly, enforcing the incompressibility constraint $\nabla \cdot \mathbf{u} = 0$:

$$\mathbf{K}_{\text{div}}(\mathbf{x}, \mathbf{x}') = \frac{\sigma^2}{\ell_s^2} \exp\left(-\frac{r^2}{2\ell_s^2}\right) \left(\mathbf{I} - \frac{(\mathbf{x} - \mathbf{x}')(\mathbf{x} - \mathbf{x}')^\top}{\ell_s^2}\right) \quad (8)$$

This kernel ensures that any sample drawn from the GP prior automatically satisfies incompressible flow, encoding a physical prior without requiring post-hoc enforcement.

4.3.3 Posterior Inference

Given training inputs $\mathbf{X}$ and stacked residuals:

$$\mathbf{y} = [r_1^u, \dots, r_N^u, r_1^v, \dots, r_N^v]^\top \quad (9)$$

the GP posterior mean and covariance at test locations $\mathbf{X}_*$ are:

$$\boldsymbol{\mu}_* = \mathbf{K}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y} \quad (10)$$

$$\boldsymbol{\Sigma}_* = \mathbf{K}_{**} - \mathbf{K}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{K}_* \quad (11)$$

where $\mathbf{K}$ is the training covariance matrix, $\mathbf{K}_*$ is the cross-covariance between training and test points, $\mathbf{K}_{**}$ is the test covariance, and σ_n^2 is the observation noise variance. The final wind field prediction at any grid point is:

$$\hat{u}(\mathbf{x}, t) = \hat{u}_{\text{GBT}}(\mathbf{x}, t) + \boldsymbol{\mu}_*^u(\mathbf{x}, t) \quad (12)$$

and equivalently for $\hat{v}(\mathbf{x}, t)$.

4.3.4 Hyperparameter Optimization

The kernel hyperparameters $\{\ell_s, \ell_t, \sigma, \sigma_n\}$ are optimized by maximizing the log marginal likelihood of the training residuals:

$$\log p(\mathbf{y} | \mathbf{X}) = -\frac{1}{2} \mathbf{y}^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y} - \frac{1}{2} \log |\mathbf{K} + \sigma_n^2 \mathbf{I}| - \frac{n}{2} \log 2\pi \quad (13)$$

Optimization is performed using L-BFGS-B with 5 random restarts to avoid local minima, with physically motivated bounds on ℓ_s (0.5 to 3.0 degrees, corresponding to approximately 50–300 km) and ℓ_t (0.1 to 0.9 normalized time units). To keep the GP covariance matrix tractable, training points are subsampled to every 24th timestamp, yielding 700 spatiotemporal training points (100 stations $\times$ 7 timestamps) and a covariance matrix of size 1400×1400 .

4.4 Baselines

Two baselines are used for comparison. The **spatial average** predicts the global mean of all station observations at every held-out point, representing the simplest possible estimator with no spatial reasoning. **Ordinary kriging** Cressie [1993] fits a spherical variogram to the station observations and interpolates spatially using weighted distance, representing the classical geostatistical approach to spatial wind interpolation.

4.5 Evaluation Metrics

The pipeline is evaluated on held-out ERA5 grid points using two metrics. Root mean squared error (RMSE) measures the accuracy of the mean prediction:

$$\text{RMSE} = \sqrt{\frac{1}{M} \sum_{j=1}^M (\hat{u}_j - u_j)^2} \quad (14)$$

Coverage measures uncertainty calibration, specifically the fraction of held-out points where the true value falls within the GP's 95% posterior confidence interval:

$$\text{Coverage} = \frac{1}{M} \sum_{j=1}^M \mathbf{1}[\hat{u}_j - 1.96 \sigma_j \leq u_j \leq \hat{u}_j + 1.96 \sigma_j] \quad (15)$$

A well-calibrated model achieves coverage close to 95%.

5 Results & Analysis

5.1 Experimental Setup

Experiments are conducted on ERA5 reanalysis data over southern California (32–38°N, 115–122°W) at 850 hPa pressure level, covering January 1–7, 2020 at hourly resolution. The dataset consists of 725 grid points across 168 timestamps, yielding 121,800 total spatiotemporal observations. A random subset of 100 grid points are designated as station locations, fixed across all timestamps, giving 16,800 training rows for the GBT and 105,000 held-out rows for evaluation. The GP is trained on a subsampled set of 700 points (100 stations $\times$ 7 timestamps, every 24th timestamp) for computational tractability.

The data is split by location rather than timestamp — 100 grid points are randomly designated as stations (fixed across all timestamps) and the remaining 625 locations are held out for evaluation. This simulates a realistic scenario where a fixed network of weather stations is used to estimate wind at unobserved locations. All GBT features are standardized using a StandardScaler fitted on training data only. The GBT uses 200 estimators, a learning rate of 0.05, and maximum depth of 4. GP hyperparameters are optimized via L-BFGS-B with 5 random restarts on the log marginal likelihood.

5.2 Stage 1: Gradient Boosted Trees

The GBT achieves a baseline RMSE of 2.59 m/s for u and 3.16 m/s for v on held-out grid points. Feature importances for the u model are shown in Table 1. Geopotential is the most important feature (0.381), reflecting the strong influence of terrain height on 850 hPa wind over the complex topography of southern California. Temperature ranks second (0.190), followed by latitude (0.179). The cyclic hour features contribute relatively little (0.066 each), suggesting that over a one-week period, day-to-day synoptic variability dominates the diurnal cycle.

Table 1: GBT feature importances for the u wind component model.

Feature	Importance
Geopotential (z)	0.381
Temperature (T)	0.190
Latitude	0.179
Longitude	0.118
Hour (sin)	0.066
Hour (cos)	0.066

5.3 Stage 2: Gaussian Process

Table 2 summarizes RMSE and coverage results for all methods. The optimized hyperparameters for each kernel are reported in Table 3.

Table 2: RMSE and coverage comparison across all methods. Coverage is the fraction of held-out points where the true value falls within the GP’s 95% confidence interval. Baselines are evaluated on a random subset of 5,000 held-out points.

Method	RMSE- u (m/s)	RMSE- v (m/s)	Coverage- u	Coverage- v
Spatial Average	3.36	5.72	—	—
Kriging	3.56	5.74	—	—
GBT only	2.59	3.16	—	—
GBT + RBF ST	1.81	2.19	90.75%	83.90%
GBT + Matérn 5/2 ST	1.73	2.13	91.49%	84.38%
GBT + Divergence-Free ST	2.23	2.71	73.49%	68.10%

Table 3: Optimized hyperparameters for each GP kernel.

Kernel	ℓ_s (deg)	ℓ_t	σ	σ_n
RBF ST	0.799	0.169	2.450	0.607
Matérn 5/2 ST	0.999	0.169	2.429	0.607
Divergence-Free ST	1.358	0.211	4.246	1.000

5.4 Analysis

The spatiotemporal GP consistently improves over GBT alone across all kernel variants except divergence-free. The best performing method, GBT + Matérn 5/2 ST, achieves RMSE of 1.73 m/s for u and 2.13 m/s for v , representing a 33% and 33% reduction respectively over GBT alone. Both RBF and Matérn kernels achieve coverage close to the 95% target (90.75% and 91.49% for u respectively), indicating well-calibrated uncertainty estimates.

The spatial average and kriging baselines perform significantly worse, with RMSE above 3.3 m/s for u and above 5.7 m/s for v . Notably, kriging performs no better than the spatial average, which can be attributed to the purely spatial nature of kriging being evaluated against a temporally varying test set spanning 168 timestamps.

The divergence-free kernel underperforms relative to RBF and Matérn on RMSE, achieving 2.23 m/s for u compared to 1.73 m/s for Matérn 5/2. This is most likely attributed to the incompressibility constraint being an imperfect prior for 850 hPa winds over complex terrain, where orographic effects and boundary layer interactions introduce apparent divergence in the flow. However, the divergence-free kernel achieves 73.49% coverage, a substantial improvement over the single-snapshot results obtained during development, suggesting the temporal dimension helps it express more meaningful uncertainty.

5.5 Qualitative Results

Figure 1 shows the reconstructed wind field at hour 20:00 UTC on January 1, 2020 (noon in southern California). The predicted field closely matches the ERA5 ground truth in station-dense regions, while the uncertainty panel correctly identifies sparse regions with elevated posterior variance. A localized artifact is visible near 35°N, 119.5°W, where the absence of nearby stations causes the GP residual correction to overcorrect, resulting in reduced wind speed. This is correctly flagged as a high-uncertainty region by the posterior variance, demonstrating that the model appropriately identifies its own limitations.

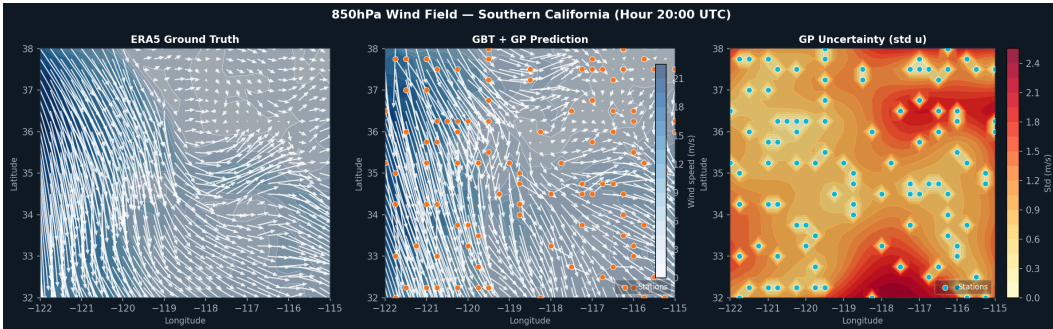


Figure 1: 850 hPa wind field reconstruction at 20:00 UTC, January 1, 2020. Left: ERA5 ground truth. Middle: GBT + GP mean prediction with station locations shown as orange dots. Right: GP posterior standard deviation for u , with station locations shown as cyan dots. Uncertainty is lowest near stations and highest in data-sparse regions.

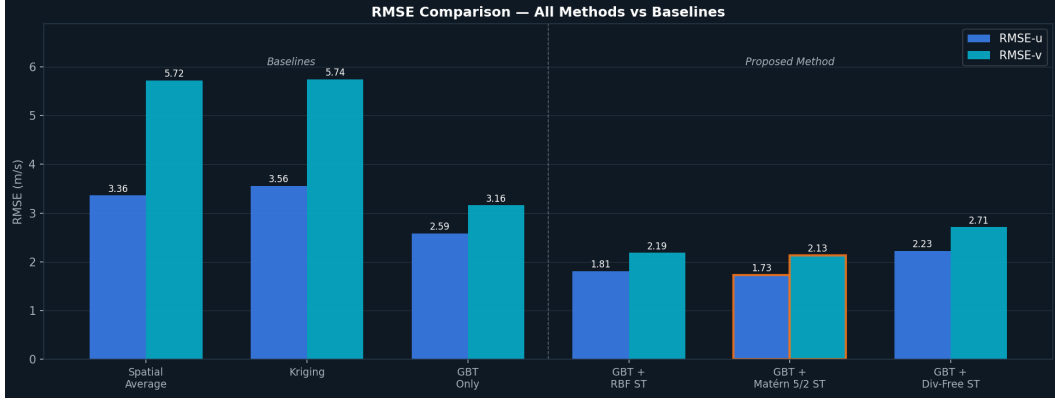


Figure 2: RMSE comparison across all methods for u and v wind components. The proposed GBT + Matérn 5/2 ST method (highlighted) achieves the lowest RMSE, outperforming all baselines and kernel variants.

6 Conclusion

This project proposed a two-stage pipeline combining Gradient Boosted Trees and physics-informed Gaussian Process regression for spatiotemporal wind field estimation at 850 hPa. The GBT stage captures nonlinear relationships between local meteorological features and wind speed, while the GP stage interpolates the residuals spatially using a spatiotemporal kernel with learned hyperparameters. The pipeline was evaluated on ERA5 reanalysis data over southern California across a one-week period, simulating a realistic sparse station network relevant to UAV-based cloud seeding operations.

The proposed method achieves a 33% RMSE reduction over GBT alone and 91% uncertainty coverage using the Matérn 5/2 spatiotemporal kernel, significantly outperforming spatial average and kriging baselines. The well-calibrated uncertainty estimates are particularly valuable for mission-critical applications such as cloud seeding, where knowing the confidence of a wind estimate is as important as the estimate itself. Qualitative analysis of the reconstructed wind field demonstrates that the model performs well in station-dense regions and correctly flags high uncertainty in data-sparse areas, including a localized artifact near 35°N , 119.5°W that the posterior variance correctly identifies as unreliable. More broadly, this project demonstrates that embedding physical structure into the covariance kernel of a GP is a principled and effective way to combine domain knowledge with data-driven learning, offering a middle ground between expensive physics-based simulation and machine learning.

The divergence-free kernel, while theoretically well-motivated by the incompressibility of atmospheric flow, underperforms empirically relative to Matérn 5/2. This is attributed to orographic effects and boundary layer interactions at 850 hPa over complex terrain introducing apparent divergence that violates the kernel’s physical prior. This finding suggests that the divergence-free kernel may be better suited to smoother flow regimes such as upper tropospheric winds or open ocean domains.

Several limitations of this work point to directions for future investigation. First, the GP is trained on a subsampled set of 700 spatiotemporal points for computational tractability; sparse GP approximations using inducing points could enable scaling to the full dataset without subsampling. Second, uncertainty calibration was evaluated only through coverage, and a full calibration curve across multiple confidence levels would provide a more complete picture. Third, the current pipeline treats each wind component independently in the GBT stage; a joint vector-valued formulation throughout could better capture the coupling between u and v . Finally, evaluation across multiple geographic regions and weather regimes would strengthen the generalizability of the conclusions.

7 Code & GitHub

The code for this project is available at https://github.com/Niklichang/ECE228_Final_Project.

8 Team Member Contribution

Niklas Chang (me) is the sole contributor to this project.

Bonus: Course Evaluation Submission

Evaluation Form

Note
Thank you for submitting your evaluation. You may edit and re-submit it at any time prior to the deadline.

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Yuanyuan Shi

Primary Instructor for

ECE 228 - ML for Physical Applications [A00] (Shi, Yuanyuan) - SP26

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Student Learning

Figure 3: Your caption here

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